Work Breakdown Structure (WBS): Kaudi Game

Project Name: Kaudi – Cowrie Shell Scoring Game & Shared Collection Dice Engine
Institution: Centurion University of Technology and Management (CUTM)
Author / Lead Developer: Game Development Studio & Engineering Team
Collection: Native & Indigenous Sports of India (Title 18)

Level	WBS Code	Element Name (Development Steps)	Definition (Clearly define work to be done)
1	**1.0**	**Project Startup & Framework Initialization**	Initial project scoping, technical requirements, repo initialization, and standard setup.
2	1.1	Game Concept & Rules Formalization	Extract authentic rules from CUTM Indigenous Sports Anthology (2025); establish standard scoring matrix for 4, 6, and 7 cowrie shell configurations.
2	1.2	Architecture Design & Layer Decoupling	Formalize decoupled architecture separating pure rules/RNG logic (`Core.Rules`) from renderer/canvas UI (`Runtime/UI`).
2	1.3	Version Control & Asset Pipelines	Set up Git repository, directory structures, LFS tracking for audio/visual assets, and code formatting rules.
1	**2.0**	**Core Dice Engine (Shared Collection Service)**	Authoring the deterministic, reusable Cowrie Shell simulation module used across all collection games.
2	2.1	Seeded Cowrie RNG Module	Implement cryptographically sound or seedable Pseudo-Random Number Generator (PRNG) simulating true shell physics landing distribution (60% flat mouth-up vs 40% convex dorsal).
2	2.2	Value & Special Throw Tables	Code lookup tables for standard points, all-mouth-up (Chamma), all-mouth-down (Ashtam), and individual shell point aggregations.
2	2.3	Grace Rolls & Bonus System	Implement consecutive bonus rolls logic, capping at defined limits (e.g., maximum 3 consecutive bonus turns) to prevent infinite loops.

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2	2.4	Unit Testing & Monte Carlo Validation	Execute headless automated simulations (1,000,000 rolls) to verify expected binomial distribution and bank/bust scoring balance.
1	**3.0**	**Game Mechanics & State Machine**	Implementation of player flow, game modes, scoring, and turn governance.
2	3.1	Match Finite State Machine (FSM)	Implement discrete states: `Boot`, `MainMenu`, `Setup`, `Throw`, `Resolve`, `DecideBankOrRoll`, `NextPlayer`, `GameOver`, and `Paused`.
2	3.2	Classic Target Mode Logic	Program the standard race-to-N points (default 100 points) with 2, 3, or 4 players in local pass-and-play.
2	3.3	Press-Your-Luck Risk Mode	Implement temporary round-bank accumulation where players decide whether to 'Bank' or 'Roll Again', losing round earnings if a 'Bust' combination is rolled.
2	3.4	Player Session & Turn Management	Manage active player rotations, player score tallies, lead tracking, and round summary generation.
1	**4.0**	**Artificial Intelligence (AI Opponent)**	Developing heuristic AI logic to support single-player vs. CPU gameplay.
2	4.1	Expected Value (EV) Decision Engine	Model mathematical Expected Value calculations comparing current banked score against risk-of-bust probability on next throw.
2	4.2	Dynamic Race Context Adaptation	Implement risk modifiers: AI plays conservatively when defending a lead, but takes calculated gambles when trailing behind human players.
2	4.3	Difficulty Tier Tuning	Create 3 distinct AI profiles: Easy (conservative, banks early), Medium (pure EV threshold), Hard (race-aware aggressive pushing).
1	**5.0**	**Art, Visuals & Tactile Animation**	Asset creation and rendering in strict compliance with the Centurion University Design Bible.
2	5.1	Cowrie Shell Sprite & Normal Maps	Author high-resolution authentic 2D sprites (128x128px) depicting natural cowrie shells in mouth-up (aperture) and mouth-down (dorsal) orientations.

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2	5.2	Toss & Scatter Animation System	Implement organic rotational velocity, trajectory arcs, multi-bounce deceleration, and settling animations for realistic shell behavior.
2	5.3	Heritage Board Mat & Tray Texture	Design warm wood board, woven coir mat, and temple sandstone textures using Design Bible tokens (#FBF4E8, #241C16, #8A5A2B, #5A3919).
2	5.4	Visual FX & Juiciness	Add celebratory particles on special rolls (Chamma / Ashtam), score floaters (+12), turn indicator pulses, and winning crown effects.
1	**6.0**	**Audio & Acoustic Design**	Authentic sound effect recording, procedural synthesis, and ambient music integration.
2	6.1	Shell Toss & Clack Audio	Produce multi-sampled acoustic recordings of real cowrie shells shaking in hand, scattering, clicking against wood, and coming to rest.
2	6.2	UI Feedback & Sting SFX	Create crisp UI button clicks, positive scoring chirps, bust thuds, turn change chimes, and victory stings.
2	6.3	Traditional Indian Ambient Music	Compose and loop subtle, relaxing background music featuring acoustic Indian folk motifs (Bansuri flute, Sitar, and soft Tabla rhythms).
2	6.4	Web Audio API Engine	Build an auto-unlocking, low-latency audio manager supporting volume sliders, sound mute toggles, and memory cleanup.
1	**7.0**	**User Interface (UI) & User Experience (UX)**	Building responsive, high-contrast, accessible presentation screens.
2	7.1	Title & Mode Selection Screens	Build clean entrance menu with options for 1P vs AI, 2P Local, 4P Local, Target Score adjustments (50, 100, 150), and Shell Count (4, 6, 7).
2	7.2	In-Game HUD & Scoreboard	Position player status cards, target progress bars, current roll readout, and prominent minimum 88x88px action buttons (Toss, Bank).
2	7.3	Cultural Lore & How-to-Play Modal	Integrate an illustrated pedagogical carousel explaining the history of Kaudi in Indian antiquity, shell morphology, and scoring rules.
2	7.4	Responsive Layout &	Implement flexible CSS/canvas scaling

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		Accessibility	supporting portrait phones, tablets, and 16:9 widescreen monitors with WCAG AAA contrast ratios.
1	**8.0**	**Quality Assurance & Deployment**	Verification, device compatibility testing, and multi-channel publication.
2	8.1	Cross-Browser & Device Matrix QA	Test touch response, audio unlock, and rendering across Chrome, Safari (iOS/macOS), Firefox, Edge, and Android WebView.
2	8.2	WebGL & PWA Packaging	Generate production builds, Service Worker caching for 100% offline play, Web App Manifest, and Brotli compression headers.
2	8.3	Final Store & Portal Publishing	Host on CUTM Heritage Game Portal, prepare itch.io page, and package APK/AAB for Google Play distribution.